

Translation/Dialogue Tutorial: Statistical Methods for Fairness in Machine Learning

Momin M. Malik Dave Watson Deepak Sharma Mahmoud AlJuhani
Sunghwan Sohn Elif Polat Chung-Il Wi Young J. Juhn

Mayo Clinic

June 27, 2026

Presented at the 2026 ACM Conference on Fairness, Accountability, and Transparency (FAccT '26)
Montréal, Quebec, Canada, June 25–28, 2026

Slides at <https://www.mominmalik.com/facct2026.pdf>

Outline

Introduction

Statistical background

Recommended intervals

Power and sample size

Demo

Other Topics

Conclusion

References

Outline

Introduction

Statistical background

Recommended intervals

Power and sample size

Demo

Other Topics

Conclusion

References

Three-slide summary 1/3: Guiding recommendations

- Technical aspects are all subsequent to system purpose and goals, power imbalances, audit study designs, actionability, and accountability.
- Statistical methods won't make your model "fair". But given a quantitative definition, they are the right way to measure if your model is "fair" or not.
- Quantify uncertainty! Test whether the observed differences/ratios are within what we would expect from natural variability for a given sample size, or greater.
- The methods covered here are better than bootstrapping, and even better than "exact" intervals (distributions are choppy for small n , smoothing out works better).
- "Data mining" for bias runs into the multiple comparisons problem; adjust for this, or better yet, substantively decide on groups/metrics to compare (especially for intersections of group membership!)
- Heed the limits of quantification.

Three-slide summary 2/3: Methodological recommendations

- Model selection: CIs over k folds okay (e.g., same performance but more “fair”).
- “Fairness” claims: on completely held-out test set (Bates et al., 2023), calculate point estimates *and* CIs (or a p -value, but $H_0 \notin \text{CI}_{95\%} \iff p < 0.05$).
- Goal: demonstrate “statistical **equivalence**” (or non-inferiority) within effect size range, not just “statistically significant difference”
- Multiple comparisons: avoid, or else Bonferroni correction (or FDR).
- Check if sample size of smaller group gives sufficient statistical power.

Three-slide summary 3/3: Technique recommendations

- Binomial proportions (e.g., accuracy, precision, recall): **Wilson score** intervals. Big n : Agresti-Coull also okay.
 - Paired data (e.g., model comparisons) with **McNemar** test.
- Difference in binomial proportions: **Agresti-Min exact unconditional** interval for small/medium n ; for medium/large, **Newcombe hybrid score** interval.
- Ratios of binomial proportions (“risk ratios” or “relative risk”: e.g., “equal opportunity”, “predictive parity”, etc.): **Koopman asymptotic score** interval.
- Odds ratios: **Baptista-Pike mid- p** interval.
 - Mathematically, when $\hat{p} \gg 0$, odds ratios are better.
- All in `contingencytables` in R. Formulas in Fagerland et al. (2017).

Learning goals

1. Know the applicable statistical best practices for making claims around standard ML fairness metrics
2. Understand something about why those are best practices
3. Be able to use relevant software packages to calculate confidence intervals
4. Use dotcharts or forest plots to visualize confidence intervals
5. Know the relevance of doing power/sample size calculations
6. Be able to carry out power/sample size calculations

Context of the material covered here

These methods are what machine learning should have been using all along! Yes, things are messy, but they have been investigated thoroughly over more than a century. The methods are fully developed and tested, and implemented in software, so we should start using them!

E.g., AI Fairness 360, TensorFlow Fairness Indicators, fairmodels, fairlearn, Aequitus, etc., can and should be using these methods. Currently, none of them are.

Even existing statistical work in FAccT (Lum et al., [2022](#)), statistical theory in actuarial science (Aigner et al., [2023](#)), and actuarial science practice (Relman Colfax PLLC, [2021](#), [2022](#)) does not use or cover these known analytic methods for inference.

Context of this presentation

- Assumed background: graduate-level probability and mathematical statistics course (e.g., Casella & Berger, or *All of Statistics*).
 - Less background: you can still use the techniques and “turn the ‘correct’ crank”
- Pace: There is a LOT of content that I’ll go through quickly; my goal is to give pointers.
- Slides are at <https://www.mominmalik.com/facct2026.pdf>, but without images from Fagerland et al. (2017) since I didn’t seek permission

Disclosures: None.

This work is supported by National Institute of Health (NIH)-funded R01 grant [R01HL171508-01A1](#).

Caveats 1/2

- How much do people care about uncertainty quantification? How much does it actually matter? How much can we actually use it?
 - I'm not sure. I'm looking into it. Perpetual gripe for statisticians (whether justified or not) that people ignore uncertainty, and go with point estimates (or significance cutoffs at 0.05).
- How much does it matter in practice to use the best methods, over simpler ones?
 - Also, not sure. Just found out from Miranda Bogen and Logan Koepke (Black et al., 2026): in credit scoring, they apparently just use t -tests (Relman Colfax PLLC, 2021, 2022)!
 - Even if it did make a difference, would it matter in, say, a court of law? Would it ever lead to different decisions about system design, adoption, etc.?
 - Unclear; but, being able to use better and deeper methods might help other goals

Caveats 2/2

- Is this the right thing to be spending *any* time on? Or, how much?
 - Certainly not the most important thing. Theory of change: use this as an entry point into more important discussions.
 - How much time: hopefully next to none, since the work is already done. Still, time and effort in explaining to others and getting them on board, training people in how to adopt, implementing in larger systems
- What about model multiplicity?
 - This is not estimating anything about the *world*—it is estimating something about a given model. It is a “bivariate” assessment, of a single dichotomous covariate against a binary label (potentially testing multiple dichotomous covariates, and constructing new ones as intersections).
 - However, if you do assessment with a logistic regression to adjust for multiple things at once, then your model specification (your “risk adjustment”) will matter

How this work came to be

- In graduate coursework, I learned “optimism”, the amount by which training data overestimates performance versus test data. *Training performance is a statistically biased estimator of out-of-sample performance.*
- Machine learning performance claims are really statistical estimators. Regardless of *model's* bias-variance tradeoff, and ignoring inference on model weights, we want unbiased estimates of and statistical inference for *performance* (Kapoor et al., 2024; Malik, 2020)
- I looked for asymptotic distributions of precision, recall, etc., without success (I later realized they are just binomial!)
- Ratios of precision, recall, etc.: convolutions of two binomial distributions? Convolutions of marginals/conditionals of a bivariate binomial??...
- From biostatisticians, I learned that there is a fully-formed framework within *diagnostic testing*. E.g., “fairness” metrics are risk ratios, and CIs exist for those!

Outline

Introduction

Statistical background

Recommended intervals

Power and sample size

Demo

Other Topics

Conclusion

References

Review: Bernoulli distribution

Estimated probabilities, from successes out of a total (e.g., precision is True Positives out of Predicted Positives; Recall is True Positives out of Real Positives) induce a Bernoulli distribution, a special form of the Binomial distribution. Minimally:

$$\Pr(k; p) = \begin{cases} p & \text{if } k = 1 \\ 1 - p & \text{if } k = 0 \end{cases}$$

Easier to use parameterizations. With $k \in \{0, 1\}$, $\Pr(k; p) = pk + (1 - p)(1 - k)$ OR $\Pr(k; p) = p^k(1 - p)^{(1-k)}$. For $X \sim \text{Bernoulli}(p)$, $E(X) = p^1(1 - p)^0 = p$, and

$$\text{Var}(X) = E(X^2) - E(X)^2 = (p \cdot 1^2 + (1 - p) \cdot 0^2) - p^2 = p(1 - p).$$

Unlike a Gaussian, with its separate parameter σ^2 for the variance, Binomial variance is a function of the mean. And, it is maximized when $p = 0.5$. This contributes to issues.

Review: Binomial proportion confidence intervals

- Normal-based (Wald) approximation

$$\hat{p} \pm z_{\alpha} \sqrt{\hat{p}(1 - \hat{p})/n}$$

has “zero-width intervals” for $0 \leftarrow \hat{p} \rightarrow 1$; “overshoots” when $\hat{p} < 1/(1 + n/z_{\alpha}^2)$.

- Clopper-Pearson interval does out the Binomial combinatorics exactly. CI expressible as quantiles of an F -distribution. In practice, covers more than 95% (next slide)
- Best-performing alternative is still the 1927 Wilson score interval. Approximates the Bernoulli as Gaussian, leading to some algebra interpretable as pseudocounts.

$$\frac{1}{1 + z_{\alpha}^2/n} \left[\left(\hat{p} + \frac{z_{\alpha}^2}{2n} \right) \pm \frac{z_{\alpha}}{2n} \sqrt{4n\hat{p}(1 - \hat{p}) + z_{\alpha}^2} \right]$$

- Agresti-Coull (1998) is better for larger sample sizes.
- These, and more, implemented widely in R and Python.

Probability “coverage”

- **Coverage:** If we simulate n_{sim} datasets and calculate a 95% confidence interval (CI) on each, the interval should contain the true value for about $.95 \times n_{sim}$ of the datasets. This should be regardless of the value of p or $\hat{p}$, or n .

Figure 1: Figure 2.2 from Fagerland et al. (2017): Coverage probabilities of three confidence intervals for the binomial probability.

Aside: Types of intervals

- **Exact:** Sum exact probabilities of all contingency tables that agree less than or equally with the null than does the observed table. Varieties: Exact conditional and exact unconditional (superior)
- **Wald:** Normal approximations
- **Score:** Take the derivative of the likelihood, and evaluate at the null hypothesis
- **Mid- p :** Sum exact probabilities of all contingency tables that agree less than the null than does the observed table, plus half the probability of the ones that agree equally. “Although the approach... seems like an ad-hoc solution, it has a solid theoretical justification.”
- Other types of intervals: Likelihood Ratio, Bayes, inverse hyperbolic sine, MOVER

Statistical equivalence and non-inferiority

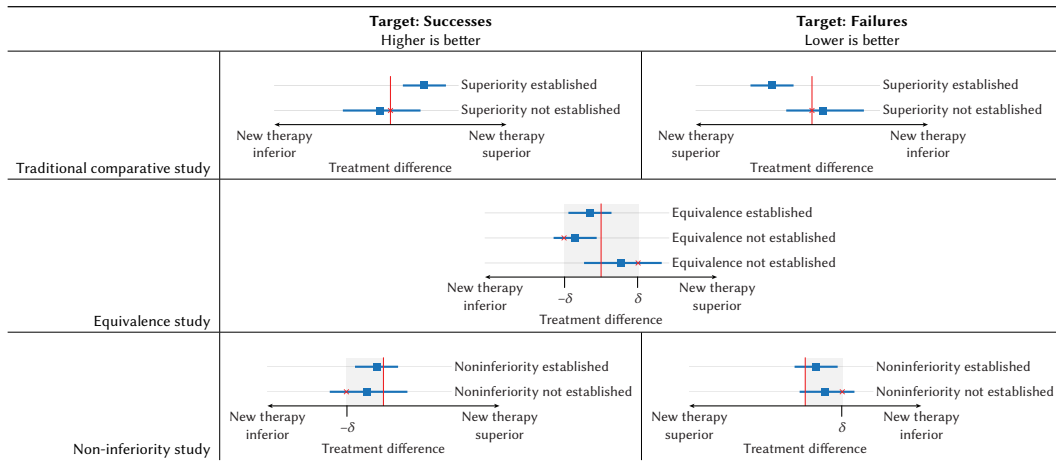


Table 1: Figure based on Walker and Nowacki (2010).

Theory of the confusion matrix

- Confusion matrix is a bivariate binomial distribution. Not useful to treat it as such, but sufficient statistic sets are of size 4 (4 cell counts; N and three cell counts/fractions; Precision, Recall, Prevalence, and N ; etc.)

Real	Predicted		Total		
	Positive	Negative			
Positive	TP	FN	RP	→	TPR
Negative	FP	TN	RN	→	TNR
Total	PP	PN	N		
	↓	↓		→	
	PPV	NPV			Accuracy

- Row conditionals: True Positive Rate [recall, sensitivity], True Negative Rate [specificity]. Complements in $[0, 1]$: False Negative Rate, False Positive Rate.
- Column conditionals: Positive Predictive Value [precision], Negative Predictive Value. Complements in $[0, 1]$: False Discovery Rate, False Omission Rate.
- Diagonal is Accuracy; complement (off-diagonal) is Error Rate.

Bayes rule connecting rows to columns with prevalence

- PPV (precision) $\propto$ TPR (recall, sensitivity) $\times$ prevalence (base rate), normalized by TPR, TNR (specificity), and prevalence.

$$\begin{aligned}\text{PPV} &= \Pr(Y = 1 | \hat{Y} = 1) \\ &= \frac{\Pr(\hat{Y} = 1 | Y = 1) \cdot \Pr(Y = 1)}{\Pr(\hat{Y} = 1 | Y = 1) \cdot \Pr(Y = 1) + \Pr(\hat{Y} = 1 | Y = 0) \cdot \Pr(Y = 0)} \\ &= \frac{\text{TPR} \cdot \text{Prevalence}}{\text{TPR} \cdot \text{Prevalence} + (1 - \text{TNR}) \cdot (1 - \text{Prevalence})}\end{aligned}$$

$$\begin{aligned}\text{NPV} &= \Pr(Y = 0 | \hat{Y} = 0) \\ &= \frac{\Pr(\hat{Y} = 0 | Y = 0) \cdot \Pr(Y = 0)}{\Pr(\hat{Y} = 0 | Y = 1) \cdot \Pr(Y = 1) + \Pr(\hat{Y} = 0 | Y = 0) \cdot \Pr(Y = 0)} \\ &= \frac{\text{TNR} \cdot \text{Prevalence}}{\text{TPR} \cdot \text{Prevalence} + (1 - \text{TNR}) \cdot (1 - \text{Prevalence})}\end{aligned}$$

Diagnostic testing and case-control data

- Diagnostic testing in medicine is like machine learning: there is a gold standard test ($\iff$ “ground truth”) that is expensive, onerous, or not available in advance. There is some cheaper, easier, or earlier proxy test ($\iff$ classifier).
- In medicine, only know the labels of tested patients. Biostats makes heavy use of “case-control” data. Cases = confirmed positives, controls = confirmed negatives. Sampling done by a chosen ratio, like 1:1, or 10:1.
- This is a type of sampling on the dependent variable!
- We can still estimate TPR and TNR (sensitivity and specificity), which don’t depend on prevalence (but can still have “spectrum bias”). These are “scientifically relevant”. Doctors want PPV (precision) and NPV, which are “clinically relevant”
- Classic example of Bayes rule: what is the probability that a patient is HIV positive, given a positive test with 99% sensitivity? May be less than 1%.
- (Sensitivity and specificity can have “spectrum bias”)

Side lesson: Working with medicine

- Biostats standard of case-control data, and ML standard of PPV and TPR (precision and recall), clash!
- Best is “community screening” data. Complete sample, so represents prevalence and “spectrum” of disease. But expensive to screen everyone, regardless of background risk, so seldom done.
- If given case-control data, do *not* report precision from the test set. Only report TPR and TNR.
- Need true out-of-sample testing of a classifier (“silent evaluation”) to calculate PPV.
- “Pragmatic clinical trials”: randomly assign patients to two conditions, and check diagnosis rate within each condition. Gives *only* PPV.
- These lessons also applies to other cases of non-random data.

Functions of binomial proportions

There are three relevant functions of two binomial proportions, X_1 and X_2 .

- Difference of binomial proportions, $X_1 - X_2$
 - Differences work well, but don't normalize for relative size.
 - In actuarial science, *standardized mean difference* (SMD): divide mean difference by reference ("control") group's s.d. (Glass' δ), or pooled s.d. (Cohen's d)
- Ratio of binomial proportions, X_1/X_2 .
 - In biostatistics, these are called *risk ratios* or *relative risk*.
 - This is what is standard in machine learning "fairness", e.g., as predictive parity, equal opportunity, etc.
 - In insurance, the risk ratio is called a "disparity ratio" or "adverse impact ratio" (AIR). These really are the same thing as risk ratios (Aigner et al., 2023)
- Odds ratio, $\left(\frac{X_1}{1-X_1}\right) / \left(\frac{X_2}{1-X_2}\right)$

Aside: paired (dependent) and unpaired (independent) binomial proportions

- Functions can be calculated for either *paired* data (e.g., which of two models works better for the same set of observations?) or *unpaired* data (e.g., does a model work less well on one group than another group?).
- Separate best methods for each case. Fagerland et al. (2017) has both.
- *Functions* of paired binomial proportions are less relevant for ML fairness
- Variations of McNemar's test work best for comparing paired binomial proportions
- Testing paired binomial proportions is appropriate for comparing two classifiers—not for model selection, for which k -fold is fine, but for “improvement over baseline” claims (should also correct for multiple comparisons!).
- I will focus on functions of unpaired binomial proportions, which is the fairness setting (e.g., comparing performance on two separate sets of observations)

Advantages/disadvantages: Risk ratios

- + What we “naturally” do. “The murder rate of the US (5.763 per 100K) is more than ten times than that of China (0.502 per 100K)” is a risk ratio.
- Only work well for small p . For example, if one rate is 0.5, the other rate can be, at most, twice as high.
- Complements have asymmetric CIs. E.g., CI for risk ratio of TPR may be smaller than CI for risk ratio of $(1 - \text{TPR}) = \text{FNR}$. Leads to undesirable property of difference inferences/needed sample size for two otherwise equivalent quantities

Advantages/disadvantages: Odds ratios

- + Scale well across $[0, 1]$. A ten-fold increase in the odds can happen whether the base probability is 0.0005 or 0.5.
- + Log odds ratios, via logit : $[0, 1] \rightarrow [-\infty, \infty]$, are approximately normal
- + CIs are symmetric between complementary metrics: no difference in inference on TPR and on FNR
- “Odds” are strange (unless in gambling/sports betting)
- Four-fifths rule is “epistemic trespass” (Watkins and Chen, 2024), but also, doesn’t translate into odds ratios:
 - $p_1 = .16$ and $p_2 = .2$: $RR = 0.8$, but $OR = [.16/(1 - .16)]/ [.2/(1 - .2)] \approx .762$.
 - $p_1 = .6$ and $p_2 = .75$: $RR = 0.8$ but $OR = [.6/(1 - .6)]/ [.75/(1 - .75)] = .5$.
 - We should figure out how to set meaningful effect sizes anyway

Outline

Introduction

Statistical background

Recommended intervals

Power and sample size

Demo

Other Topics

Conclusion

References

Notation and setup for comparisons between groups

To put group comparisons also within a framework of contingency tables, Fagerland et al. (2011, 2017) take the individual confusion matrices of group 1 and group 2, take the same row or column of each, and row-bind them into a new contingency table. Using their definitions:

	Successes	Failures	Sum
Group 1	n_{11}	n_{12}	n_{1+}
Group 2	n_{21}	n_{22}	n_{2+}
Sum	n_{+1}	n_{+1}	N

I.e., for PPV (precision), n_{11} is True Positives in group 1 and n_{12} is False Positives in group 1, so n_{1+} is the Predicted Positives in group 1, etc.

Convention: put the marginalized (“treated”) group the top, and the dominant/reference (“control”) group on the bottom.

First-pass confidence intervals

- Difference in proportions: Wald interval (normal-based)

$$(\hat{p}_1 - \hat{p}_2) \pm z_{\alpha/2} \sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_{1+}} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_{2+}}}$$

- Risk ratio: Katz log interval (via Delta method)

$$\log \left(\frac{\hat{p}_1}{\hat{p}_2} \right) \pm z_{\alpha/2} \sqrt{\left(\frac{1}{n_{11}} + \frac{1}{n_{21}} \right) - \left(\frac{1}{n_{1+}} + \frac{1}{n_{2+}} \right)}$$

- Odds ratio: Woolf logit interval (via Delta method)

$$\log \left(\frac{\hat{p}_1/(1-\hat{p}_1)}{\hat{p}_2/(1-\hat{p}_2)} \right) \pm z_{\alpha/2} \sqrt{\frac{1}{n_{11}} + \frac{1}{n_{12}} + \frac{1}{n_{21}} + \frac{1}{n_{22}}}$$

- All of these are fine when $\min(n_{1+}, n_{2+}) \geq 200$

Recommended intervals

- Difference in proportions:
 - Agresti-Min exact unconditional interval for “small” n , approximately $\min(n_{1+}, n_{2+}) < 50$
 - Newcombe hybrid score interval for “medium” n , approximately $50 \leq \min(n_{1+}, n_{2+}) < 200$.
Are simpler intervals, but this is more widely implemented
- Risk ratio: Koopman asymptotic score interval (next slide).
 - Other options (inverse sinh and MOVER-R Wilson) are slightly better, but Koopman is widely implemented
- Odds ratio: Baptista-Pike mid- p interval.

Example: Koopman asymptotic score interval

Let n_{11} and n_{21} respectively be the number of ‘successes’ (e.g., the overall correct labels for accuracy; the true positives for recall or precision) for the first and second group; n_{1+} and n_{2+} respectively be the counts of ‘totals’ (e.g., total group size for accuracy; actual positives for recall; predicted positives for precision); $\phi = p_1/p_2$ (for which we plug in $\hat{\phi} = \hat{p}_1/\hat{p}_2$); and

$$\tilde{p}_1 = \frac{\phi \cdot (n_{1+} + n_{21}) + n_{11} + n_{2+} - \sqrt{[\phi \cdot (n_{1+} + n_{21}) + n_{11} + n_{2+}]^2 - 4\phi \cdot N(n_{11} + n_{21})}}{2N}. \quad (1)$$

Then, define:

$$U(\phi) = \frac{(n_{11} - n_{1+}\tilde{p}_1)^2}{n_{1+}\tilde{p}_1(1 - \tilde{p}_1)} \left\{ 1 + \frac{n_{11}(\phi - \tilde{p}_1)}{n_{2+}(1 - \tilde{p}_1)} \right\}, \quad (2)$$

Estimates of an upper and lower bounds of a confidence intervals for ϕ are two solutions for setting $U(\phi)$ equal to $\chi_{1,1-\alpha}^2$, the $1 - \alpha$ percentile of a chi-squared distribution with one degree of freedom. Also is a (long) closed-form solution.

Comparative performance of risk ratio CIs

Figure 2: Fig. 4 from Fagerland et al. (2011). Coverage probabilities for four proposed intervals for two fixed effect sizes and sample sizes.

Comparison with regression

- The techniques here are related to the coefficients of logistic regression; under respective link functions of identity, log, and logit (Fagerland et al., [2017](#), p.114) but I am unsure what intervals the standard errors are closest to. I assume it's the "first pass" intervals, so, only good for $\min(n_{1+}, n_{2+}) \geq 200$
- Advantage of doing as a regression: you can control for other factors at the same time! You can model differential performance over a multitude of attributes at once, including continuous ones
- *Can* use risk ratios via log (not logit) link function for a logistic regression or risk differences via identity link function (Naimi and Whitcomb, [2020](#)), but it may have convergence issues

Clustered data via Generalized Estimating Equations

- Frequent case: you have more than one label per unit (e.g., more than one test per patient)
- *Generalized Estimating Equations* (GEEs) are an alternative to hierarchical modeling, treating any effect of clustering as a nuisance parameter. Add-on to generalized linear models
 - They adjust standard errors from a matrix of correlation structure (implementations estimate the amount of correlation, within the structure; e.g., given AR_m for fixed m , it estimates the magnitude of the autoregression)
 - Bad if you're not sure about the correlation structure, or substantively care about it, e.g., want an intraclass correlation coefficient
- You can use GEEs with a logistic regression
- You use an identity link function to get a difference in binomial proportions, or log link function to get a risk ratio, but not sure how well this works

Outline

Introduction

Statistical background

Recommended intervals

Power and sample size

Demo

Other Topics

Conclusion

References

Background: power and sample size

- Power and sample size are seldom considered for ML: you use all the data you have, and then find out if it's "enough" or not
 - Power and sample size are only for prior to data collection (afterwards, CI tells you everything)
- When you need to go out and *collect* data, how do you know what is "enough"?
- **Power** is asking, given that there is an effect of a given size, what is the probability ($1 - \beta$) that a procedure correctly rejects the null? (β is probability of Type II error)
 - We consider this probability as a function of the sample size. What sample size do we need for a given power? Usually, target power is 0.8
- **Relevance for machine learning fairness:** before building a model, we can ask, *do we have enough observations from the smaller group for equivalence or non-inferiority?*
- Apply to intersections of marginalized group membership. Intersections well-known to quickly drain statistical power

How to calculate power under approximate normality

- For effect size θ , $H_0 : \theta = 0$, test statistic T , and critical value $z_{\alpha/2}$ (e.g., $z_{0.05/2} = \Phi(0.95) \approx 1.64$),

$$\beta = \Pr(T > z_{\alpha/2} \mid \theta \neq 0)$$

- If $T = \frac{\hat{\theta} - \theta_0}{\sqrt{\text{Var}(\hat{\theta})}} = \frac{\hat{\theta}}{\sigma/\sqrt{n}}$, but $\frac{\hat{\theta} - \theta}{\sigma/\sqrt{n}} \rightarrow \mathcal{N}(0, 1)$, then

$$\begin{aligned}\beta &= 1 - \Pr\left(\frac{\hat{\theta} - \theta}{\sigma/\sqrt{n}} \leq z_{\alpha/2} - \frac{\theta}{\sigma/\sqrt{n}}\right) \approx 1 - \Phi\left(z_{\alpha/2} - \frac{\theta}{\sigma/\sqrt{n}}\right) \\ \implies \sqrt{n} &> \frac{\sigma}{\theta} \left(z_{\alpha/2} - \Phi^{-1}(1 - \beta)\right).\end{aligned}$$

- Note: sample size n comes out of the variance estimate $\text{Var}(\hat{\theta}) = \sigma^2/n$. If possible, use some existing $\widehat{\text{Var}}(\hat{\theta}) = \hat{\sigma}^2/n$, otherwise, put in a range of possible σ
- No closed-form approximation? For a grid over ranges of θ , n , and σ , simulate each cell n_{sim} times. Under what fraction of n_{sim} do we correctly reject H_0 ?

Power curves

- 5 parameters: power β , level α , effect size θ , sample size n , and variance σ .
- Power curves:** plots of $1 - \beta$ on y -axis and n on x -axis, with a few effect sizes/variances on the same graph. At what n is $\beta \geq 0.8$? E.g., for Pearson correlation:

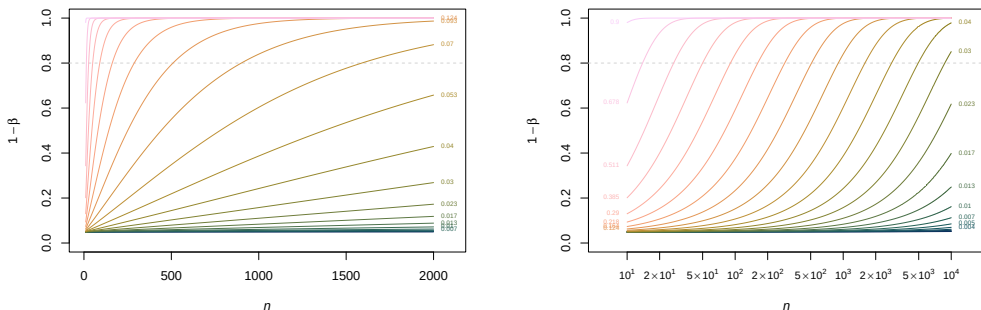


Figure 3: Power curves for correlation test at the $\alpha = 0.05$ level, in linear (left) and log scale (right). Curves are various effect sizes, $\theta = \rho$.

Power calculations for binomial proportions

- Since everything is approximate anyway, we can use the “first pass” estimates of variance from a normal approximation (plus Delta method) instead of using the more complicated intervals
- We no longer need to guess at σ or simulate over possible values, but we need an estimate of variance
 - But we have more sample size parameters: n_{11} , n_{12} , n_{21} , and n_{22} .
 - In practice, size of the smaller group $\min(n_{1+}, n_{2+})$ dominates; we will plot curves over different proportions n_{1+}/n_{2+}
- For calculating *equivalence*, for which much smaller sample sizes are okay, we incorporate the “acceptable” effect size of (0.8, 1.25), for both risk ratios and odds ratios.

Power calculations in practice

- This is one area where even professional statisticians may rely on commercial software, because it's so simple yet so tedious.
- Examples include NCSS PASS and StatXact
- There are many packages for getting power calculations for common test statistics (e.g., difference in means is most common). But if there isn't something, you can use existing functions for variance estimation and/or confidence intervals. Will show how in the demo.
- Technical note: for equivalence, we use "TOST": two one-sided tests, rather than one two-sided test

Power curves for risk ratios: Fixed effect size, across n_{1+}/n_{2+}

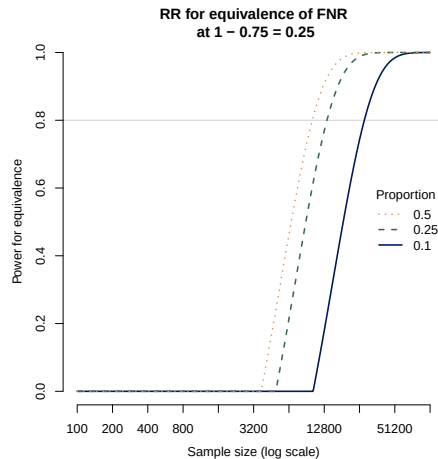
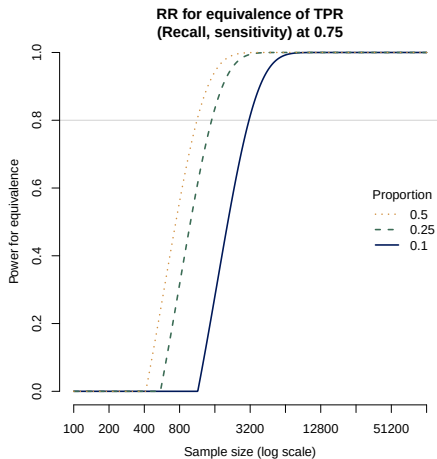


Figure 4: While $\text{TPR} = 1 - \text{FNR}$, the asymmetry of $\log^{-1} = \exp$ versus $\text{logit}^{-1} = \text{logistic}$ means that confidence intervals are asymmetric, and so sample sizes are different depending on direction.

Power curves for odds ratios: Fixed effect size, across n_{1+}/n_{2+}

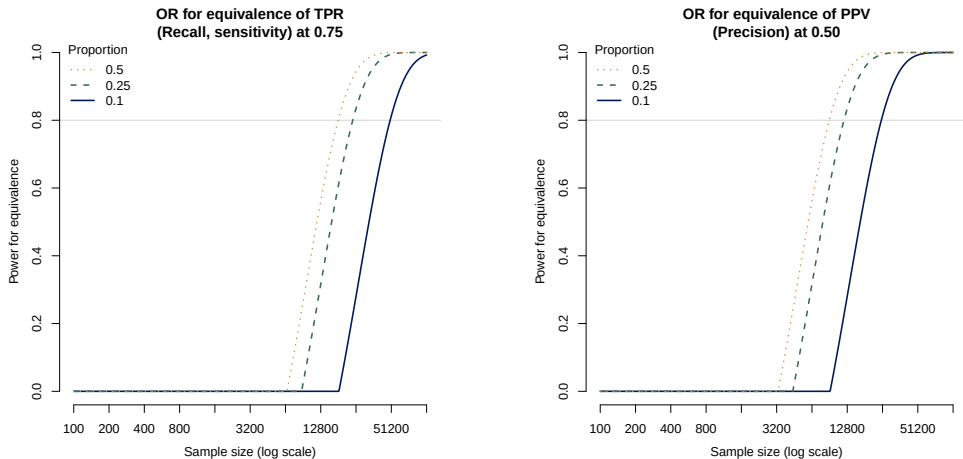


Figure 5: For odds ratios, the curves would be the exact same (not shown) for $FPR = 1 - TPR$ (left) and $FDR = 1 - PPV$ (right).

Problem for power: the multiple comparisons problem

- Try enough things, and eventually some of them will stick by chance.
 - E.g., 20 independent random normal variables gives $\binom{20}{2} = \frac{20(20-1)}{2} = 190$ pairwise comparisons. If $\alpha = 0.05$, we expect $190 \cdot 0.05 = 9.5$ p -values less than 0.05.
- Bonferroni correction is the simplest way to account for multiple comparisons. Instead of an α level, use $\alpha/\text{comparisons}$. E.g., for 190 comparisons, use $(1 - 0.05/190)$ which is $\approx 99.99974\%$ CIs instead of 95%. Simple but over-corrects; better corrections are harder.

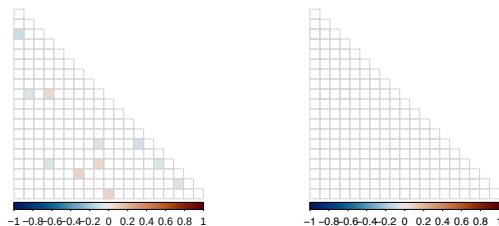


Figure 6: 20 independent (uncorrelated) random normal variables with only statistically significant coefficients shown. Left: $p < 0.05$. Right: with Bonferroni correction $p < 0.00026$.

Multiple comparisons for ML

- Machine learning deals with overfitting via cross-validation.
- If you were using CV for hypothesis testing, you'd do your data mining, then out of the things that are positive, you'd re-test only those on a new held-out folds of data. If all of those aren't still positive, then you've overfit.
- Could work out statistical properties of this (maybe it's already out there), but better to use the established statistical methods
- Post-hoc, use a Bonferroni correction; when doing power calculations, instead of level $\alpha = 0.05$, divide this by the number of comparisons you plan to make
- So, don't test every metric, on every combination of subgroup, and certainly not on every overlap of every independent subgroup categorization (in statistics, it is well known that interaction effects will sink your power). **Choose substantively!!** (Rodolfa et al., 2020)

Outline

Introduction

Statistical background

Recommended intervals

Power and sample size

Demo

Other Topics

Conclusion

References

Source and setting: previously published article from our group

- A paper from our group (Juhn et al., 2022), does things in the standard way: uses effect size cutoffs to determine results
- A set of sufficient statistics are in the paper: the number of real positives and real negatives for each group, and the precision (PPV) and recall (sensitivity) for each group

	Training cohort (N = 133)		Testing cohort (N = 113)	
	Subjects with AE (N = 34)	Subjects without AE (N = 99)	Subjects with AE (N = 40)	Subjects without AE (N = 73)
Age (in years), n (%)				
<12	28 (29.8%)	66 (70.2%)	30 (37.5%)	50 (62.5%)
≥12	6 (15.4%)	33 (84.6%)	10 (30.3%)	23 (69.7%)
Sex, n (%)				
Male	25 (32.9%)	51 (67.1%)	23 (34.3%)	44 (65.7%)
Female	9 (15.8%)	48 (84.2%)	17 (39.1%)	28 (60.9%)
Race/ethnicity, n (%)				
Non-Hispanic Whites	19 (25.0%)	57 (75.0%)	25 (37.3%)	42 (62.7%)
African Americans	5 (27.8%)	13 (72.2%)	4 (44.4%)	5 (55.6%)
Asians	2 (20.0%)	8 (80.0%)	3 (23.1%)	10 (76.9%)
Hispanics	4 (44.4%)	5 (55.6%)	3 (27.3%)	8 (72.7%)
Other categories	4 (28.6%)	10 (71.4%)	4 (33.3%)	8 (66.7%)
HOUSES, n (%)				
Q1 (the lowest SES)	6 (27.3%)	16 (72.7%)	8 (53.3%)	7 (46.7%)
Q2-Q4	23 (22.5%)	79 (77.5%)	29 (31.5%)	63 (68.5%)
Chronic condition, n (%)				
Yes	10 (33.3%)	20 (66.7%)	7 (36.8%)	12 (63.2%)
No	24 (23.3%)	79 (76.7%)	33 (35.1%)	61 (64.9%)
National ADL, n (%)				
76-100 (the lowest SES)	2 (33.3%)	4 (66.7%)	0 (0.0%)	6 (100.0%)
0-75	16 (21.1%)	60 (78.9%)	21 (32.3%)	44 (67.7%)

Figure 7: Published table 2 from Juhn et al. (2022).

Groups	Accuracy equality		Equal opportunity (sensitivity)		Predictive parity (PPV)		Predictive equality (FPR)		Balanced error rate (FPR + FNR)/2	
	NB model	GBM model	NB model	GBM model	NB model	GBM model	NB model	GBM model	NB model	GBM model
SES (HOUSES)										
Q1 (lowest SES)	0.47	0.47	0.38	0.50	0.50	0.50	0.43	0.57	0.53	0.54
Q2-Q4	0.62	0.50	0.59	0.76	0.43	0.36	0.37	0.62	0.39	0.43
Ratio (Q1/Q2-4) (1 = no diff)	0.75	0.93	0.64	0.66	1.18	1.39	1.17	0.92	1.35	1.25
Age										
<12	0.53	0.45	0.57	0.70	0.41	0.38	0.50	0.70	0.47	0.50
≥12	0.76	0.64	0.40	0.80	0.67	0.44	0.09	0.44	0.34	0.32
Ratio (<12/≥12) (1 = no diff)	0.69	0.71	1.42	0.88	0.61	0.84	5.75	1.61	1.36	1.57
Sex										
Male	0.49	0.45	0.48	0.78	0.33	0.36	0.50	0.73	0.51	0.47
Female	0.74	0.59	0.59	0.65	0.67	0.46	0.17	0.45	0.29	0.40
Ratio (male/female) (1 = no diff)	0.67	0.76	0.81	1.21	0.50	0.79	2.90	1.62	1.75	1.18
Race/Ethnicity										
Others	0.54	0.39	0.47	0.60	0.35	0.29	0.42	0.71	0.48	0.56
Non-Hispanic White	0.63	0.58	0.56	0.80	0.50	0.47	0.33	0.55	0.39	0.37
Ratio (others/White) (1 = no diff)	0.87	0.67	0.83	0.75	0.70	0.62	1.30	1.23	1.48	
Chronic condition										
At least one	0.53	0.47	0.20	0.80	0.20	0.33	0.33	0.67	0.57	0.43
None	0.61	0.50	0.59	0.69	0.46	0.39	0.38	0.60	0.39	0.46
Ratio (≥1/none) (1 = no diff)	0.87	0.94	0.34	1.16	0.43	0.86	0.88	1.11	1.44	0.95
ADI										
76-100	0.60	0.60	NC	NC	0.00	0.00	0.40	0.40	NC	NC
0-75	0.64	0.54	0.60	0.80	0.44	0.39	0.35	0.58	0.37	0.39
Ratio (76-100/0-75) (1 = no diff)	0.95	1.11	NC	NC	0.00	0.00	1.15	0.69	NC	NC

NC: not computable.

Ratios either greater than 1.2 or less than 0.8 (ie, an absolute difference between the ratio and 1 being greater than 0.2) were bolded.

Figure 8: Published table 3 from Juhn et al. (2022).

Re-analysis

Recall: we want confidence intervals that are completely contained within $(0.8, 1.25)$.

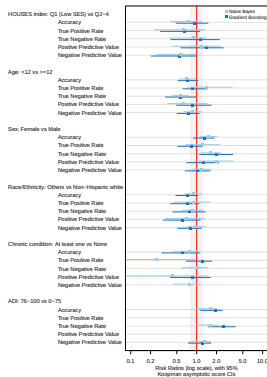


Figure 9: 60 metrics (converted to consistent directions) from (Juhn et al., 2022) plotted with (uncorrected) confidence intervals.

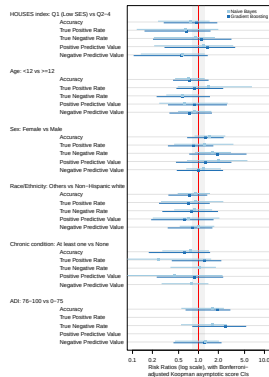


Figure 10: The same 60 metrics as on the left, with confidence intervals adj Bonferroni correction.

Demo

- Setup for using the interval calculations done by the `contingencytables` package
- A good way to visualize these: either as dot charts, or using “forest plots” (a fancier type of dot chart)
- Walkthrough of what power curve calculations look like

Outline

Introduction

Statistical background

Recommended intervals

Power and sample size

Demo

Other Topics

Conclusion

References

Non-binomial metrics

- Perhaps the most useful one that I hadn't heard about is Number Needed to Treat. This is perhaps intuitive (but it's debated). Generally, you get this from the reciprocal of a probability, and CIs follow similarly (Fagerland et al., [2011](#)).
- AUROC can be done with Wald-style intervals, but the ROC curve itself needs to be bootstrapped. Same for precision-recall curve.
- The Matthews Correlation Coefficient: reinvention of Pearson's ϕ coefficient (named by Karl Pearson in 1930s, described by Udny Yule in 1912, but first described by M. H. Doolittle in 1880s; Armistead, [2016](#)). There is a Fisher confidence interval for this.
- (Diagnostic) likelihood ratios: unsure.
- Multinomial metrics ($2 \times c$ comparisons; $r \times 2$ comparisons; and $r \times c$ comparisons) covered as well by Fagerland et al. ([2017](#)), both unpaired and ($c \times c$ only) paired.

Where do we go from here? Future work, possible extensions

- Elliptical joint confidence intervals for pairs of metrics? Certainly a row and a column with a shared cell (like precision and recall) are not independent.
- Standardized mean difference (SMD) are one thing not covered in Fagerland et al. (2017); different standardizations correspond to different named statistics, so there are probably CIs
- For clustered data, adjustments for things other than odds ratios: if use log link function (not logit) for logistic regression, get risk ratios, but has convergence issues, so may not work to apply GEEs there. Maybe can develop more direct, ad-hoc analytic approaches.
- Better understand connection between the tests described here and doing things as a logistic regression (with an identity, log, or logit link function)

Outline

Introduction

Statistical background

Recommended intervals

Power and sample size

Demo

Other Topics

Conclusion

References

Three-slide summary 1/3: Guiding recommendations

- Technical aspects are all subsequent to system purpose and goals, power imbalances, audit study designs, actionability, and accountability.
- Statistical methods won't make your model "fair". But given a quantitative definition, they are the right way to measure if your model is "fair" or not.
- Quantify uncertainty! Test whether the observed differences/ratios are within what we would expect from natural variability for a given sample size, or greater.
- The methods covered here are better than bootstrapping, and even better than "exact" intervals (distributions are choppy for small n , smoothing out works better).
- "Data mining" for bias runs into the multiple comparisons problem; adjust for this, or better yet, substantively decide on groups/metrics to compare (especially for intersections of group membership!)
- Heed the limits of quantification.

Three-slide summary 2/3: Methodological recommendations

- Model selection: CIs over k folds okay (e.g., same performance but more “fair”).
- “Fairness” claims: on completely held-out test set (Bates et al., 2023), calculate point estimates *and* CIs (or a p -value, but $H_0 \notin \text{CI}_{95\%} \iff p < 0.05$).
- Goal: demonstrate “statistical **equivalence**” (or non-inferiority) within effect size range, not just “statistically significant difference”
- Multiple comparisons: avoid, or else Bonferroni correction (or FDR).
- Check if sample size of smaller group gives sufficient statistical power.

Three-slide summary 3/3: Technique recommendations

- Binomial proportions (e.g., accuracy, precision, recall): **Wilson score** intervals. Big n : Agresti-Coull also okay.
 - Paired data (e.g., model comparisons) with **McNemar** test.
- Difference in binomial proportions: **Agresti-Min exact unconditional** interval for small/medium n ; for medium/large, **Newcombe hybrid score** interval.
- Ratios of binomial proportions (“risk ratios” or “relative risk”: e.g., “equal opportunity”, “predictive parity”, etc.): **Koopman asymptotic score** interval.
- Odds ratios: **Baptista-Pike mid- p** interval.
 - Mathematically, when $\hat{p} \gg 0$, odds ratios are better.
- All in `contingencytables` in R. Formulas in Fagerland et al. (2017).

Conclusion

- Hopefully, this is enough for you to go out and start adopting better methods!
- Or at least, you know what else is out there, to eventually work towards it.
- We are working on a tutorial paper, but this is all published anyway. Definitely cite Fagerland et al. (2017), but for now feel free to cite this tutorial as:

```
@misc{Malik2026,  
  author = {Malik, Momin M. and Watson, Dave and Sharma, Deepak and AlJuhani, Mahmoud and Sohn, Sunghwan and Polat, Elif and  
Wi, Chung-Il and Juhn, Young J.},  
  title = {Translation\slashdialogue tutorial: Statistical methods for fairness in machine learning},  
  howpublished = {Tutorial at the 2026 ACM Conference on Fairness, Accountability, and Transparency},  
  year = {2026},  
  month = jun,  
  day = {27},  
  url = {https://www.mominmalik.com/facct2026.pdf}  
}
```


Outline

Introduction

Statistical background

Recommended intervals

Power and sample size

Demo

Other Topics

Conclusion

References

References

- Aigner, D. J., del Ángel, M., & Wiles, J. (2023). Statistical approaches for assessing disparate impact in fair housing cases. *Statistics and Public Policy*, 11(1). <https://doi.org/10.1080/2330443x.2023.2263038>
- Armistead, T. W. (2016). Misunderstood and unattributed: Revisiting M. H. Doolittle's measures of association, with a note on Bayes' Theorem. *The American Statistician*, 70(1), 63–73. <https://doi.org/10.1080/00031305.2015.1086686>
- Bates, S., Hastie, T., & Tibshirani, R. (2023). Cross-validation: What does it estimate and how well does it do it? *Journal of the American Statistical Association*, 119(546), 1434–1445. <https://doi.org/10.1080/01621459.2023.2197686>
- Black, E., Bogen, M., Koepke, L., Barocas, S., Deng, W. H., & Hsu, M. (2026). The fair lending model: How the longest-running algorithmic fairness programs work in practice. *Proceedings of the 2026 ACM Conference on Fairness, Accountability, and Transparency*, 3029–3056. <https://doi.org/10.1145/3805689.3812307>
- Fagerland, M. W., Lydersen, S., & Laake, P. (2011). Recommended confidence intervals for two independent binomial proportions. *Statistical Methods in Medical Research*, 24(2), 224–254. <https://doi.org/10.1177/0962280211415469>
- Fagerland, M. W., Lydersen, S., & Laake, P. (2017). *Statistical analysis of contingency tables*. Chapman & Hall/CRC. <https://doi.org/10.1201/9781315374116>
- Juhn, Y. J., Ryu, E., Wi, C.-I., King, K. S., Malik, M., Romero-Brufau, S., Weng, C., Sohn, S., Sharp, R. R., & Halamka, J. D. (2022). Assessing socioeconomic bias in machine learning algorithms in health care: A case study of the HOUSES index. *Journal of the American Medical Informatics Association*, 29(7), 1142–1151. <https://doi.org/10.1093/jamia/ocac052>

References (cont.)

- Kapoor, S., Cantrell, E. M., Peng, K., Pham, T. H., Bail, C. A., Gundersen, O. E., Hofman, J. M., Hullman, J., Lones, M. A., Malik, M. M., Nanayakkara, P., Poldrack, R. A., Raji, I. D., Roberts, M., Salganik, M. J., Serra-Garcia, M., Stewart, B. M., Vandewiele, G., & Narayanan, A. (2024). REFORMS: Consensus-based recommendations for machine-learning-based science. *Science Advances*, 10(18), eadk3452. <https://doi.org/10.1126/sciadv.adk3452>
- Lum, K., Zhang, Y., & Bower, A. (2022). De-biasing “bias” measurement. *Proceedings of the 2022 ACM Conference on Fairness Accountability and Transparency*, 379–389. <https://doi.org/10.1145/3531146.3533105>
- Malik, M. M. (2020). A hierarchy of limitations in machine learning.
- Naimi, A. I., & Whitcomb, B. W. (2020). Estimating risk ratios and risk differences using regression. *American Journal of Epidemiology*, 189(6), 508–510. <https://doi.org/10.1093/aje/kwaa044>
- Relman Colfax PLLC. (2021). Fair lending monitorship of Upstart Network’s lending model. <https://www.relmanlaw.com/cases-upstart-network-fair-lending-counseling>
- Relman Colfax PLLC. (2022). Fair lending monitorship of Upstart Network’s lending model. <https://www.relmanlaw.com/cases-upstart-network-fair-lending-counseling>
- Rodolfa, K. T., Saleiro, P., & Ghani, R. (2020). Bias and fairness. In I. Foster, R. Ghani, R. S. Jarmin, F. Kreuter, & J. Lane (Eds.), *Big data and social science: Data science methods and tools for research and practice* (2nd, pp. 281–312). Chapman & Hall/CRC. <https://doi.org/10.1201/9780429324383-11>
- Walker, E., & Nowacki, A. S. (2010). Understanding equivalence and noninferiority testing. *Journal of General Internal Medicine*, 26(2), 192–196. <https://doi.org/10.1007/s11606-010-1513-8>

References (cont.)

Watkins, E. A., & Chen, J. (2024). The four-fifths rule is not disparate impact: A woeful tale of epistemic trespassing in algorithmic fairness. *Proceedings of the 2024 ACM Conference on Fairness, Accountability, and Transparency*, 764–775. <https://doi.org/10.1145/3630106.3658938>